

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – II

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AIM: Working with Data Sources and Data Preparation in Tableau.

- A. Connect to different data sources (Excel, CSV, Database).**
- B. Create and use data extracts vs live connectionsManage metadata (rename, hide, change data types).**
- C. Perform joins (inner, left, right) on multiple tables.**
- D. Apply data blending between two sources.**
- E. Implement different types of filters (dimension, measure, date).**
- F. Optimize performance using extracts and filtering.**

STEPS

Step 1: Import the Dataset

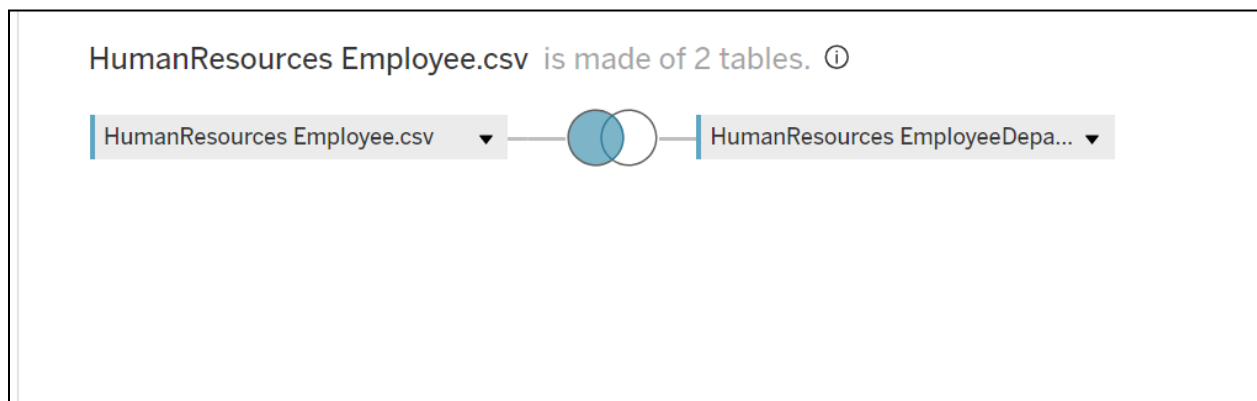
1. Open Tableau Desktop.
2. Under the Connect section, click Text File.
3. Select HumanResourcesEmployee.csv from your computer and click Open.
4. After the dataset is loaded, go to the Connection section and select Extract instead of Live.
5. Click on Sheet 1.
6. If prompted, save the extract by entering a suitable file name and click Save.
7. Save the workbook by selecting File → Save As, choose the Tableau Packaged Workbook (.twbx) format, enter a suitable file name, and click Save.

HumanResources Employee

HumanResources Emplo...

16 fields 290 rows

#	#	Abc	Abc
HumanResources Employee.csv	HumanResources Employee.csv	HumanResources Employee.csv	HumanResources Employee.csv
Business Entity ID	National ID Number	Employee Login	Organizational Level
29	948320468	adventure-works\mark1	/3/1/



Step 2: Managing Metadata

1. Open Sheet 1 and locate the required fields in the Data Pane.
2. Right-click on Job Title and select Rename. Change the field name to Employee Job Title.
3. Locate the National ID Number field, right-click on it and select Hide to remove it from the Data Pane.
4. Locate the Birth Date field and click on its data type icon.
5. If required, change the data type to Date.
6. Locate the Hire Date field and ensure that its data type is Date.
7. Verify that the metadata changes have been applied successfully.

Step 3: Performing Relational Joins on Multiple Tables

1. Click on the Data Source tab at the bottom-left corner to return to the Data Source page.
2. Double-click the HumanResourcesEmployee.csv table to open the Physical Layer.
3. From the left pane, drag the HumanResourcesEmployeeDepartmentHistory.csv file next to the HumanResourcesEmployee.csv table.
4. Tableau automatically creates a Join between the two tables.

5. Click on the Join (Venn Diagram) icon to open the Join configuration.
6. Select BusinessEntityID as the common field for both tables.
7. Change the join type and observe the results:
 - Inner Join – Displays only matching employee records available in both tables.
 - Left Join – Displays all employee records from the Employee table and matching department history records.
8. Select Left Join and keep it as the final join type.
9. Verify that the joined data is displayed correctly in the data preview.

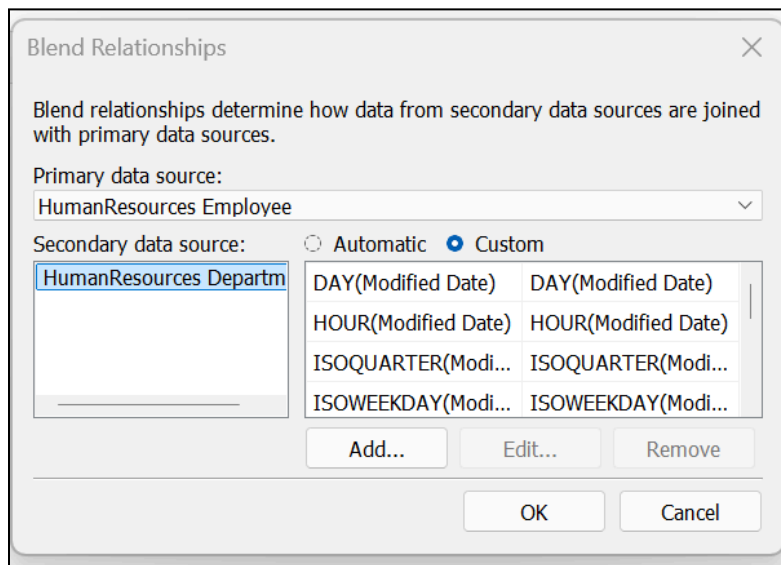
INSIGHT

- Relational joins combine employee information with department history using the BusinessEntityID field.
- The joined data provides complete employee and department assignment details.
- It improves data analysis by combining related information from multiple tables into a single dataset.

Step 4: Applying Data Blending Between Independent Sources

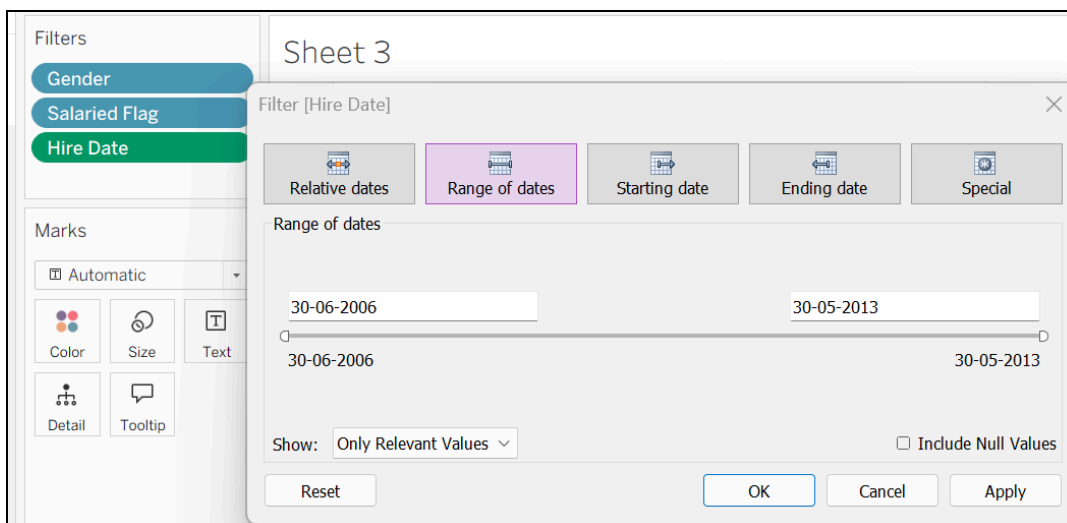
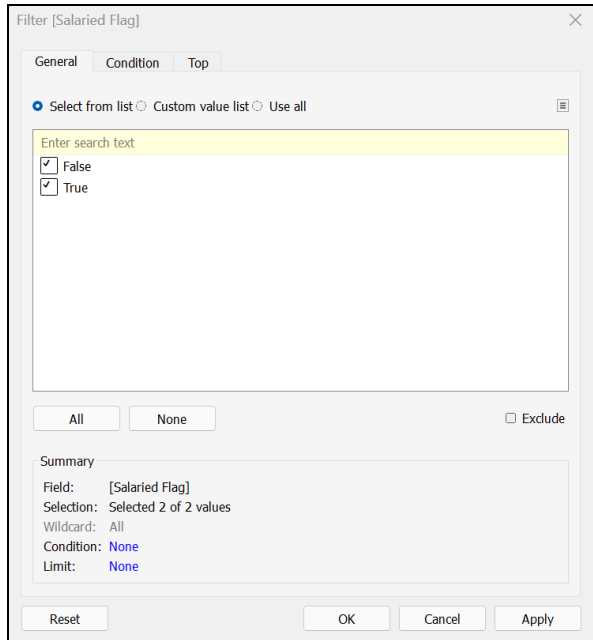
1. Click on the **New Worksheet** icon at the bottom of the Tableau window to create a new worksheet (**Sheet 2**).
2. In the **Data Pane**, ensure that the **HumanResourcesEmployee** dataset is selected as the **Primary Data Source**.
3. Drag the **Employee Job Title** field from the Employee dataset and drop it onto the **Rows** shelf. A blue check mark will appear next to the Employee data source, indicating that it has been set as the Primary Data Source.
4. From the top menu bar, click **Data** → **Edit Blend Relationships**. If the option is disabled (grayed out), click **Data** → **New Data Source**.
5. In the **Connect** pane, select **Text File** and browse to the location of **Department.csv**.
6. Select **Department.csv** and click **Open** to add it as the **Secondary Data Source**.
7. Return to **Sheet 2** and again click **Data** → **Edit Blend Relationships**.
8. In the **Edit Blend Relationships** dialog box:
 - Set **Primary Data Source** as **HumanResourcesEmployee**.
 - Set **Secondary Data Source** as **Department**.
 - Change the relationship mode from **Automatic** to **Custom**.
 - Click the **Add** button to create a custom relationship between the two datasets.
 - Select **Department ID** from the Employee dataset as the primary field.
 - Select **Department ID** from the Department dataset as the secondary field.
9. Click **OK** to save the relationship and then click **OK** again to close the dialog box.
10. Go back to the **Data Source** page and change the connection mode from **Extract** to **Live**.
11. Return to **Sheet 2**.
12. In the **Data Pane**, right-click on an empty space and select **Show Hidden Fields** so that hidden fields become visible again.

13. Locate the **Department ID** field and click the **chain link** icon beside it. The chain icon should turn **orange**, indicating that the blend relationship is active between the two data sources.
14. Select the **Department** dataset from the Data Pane.
15. Drag the **Name** (Department Name) field from the Department dataset and drop it onto the **Columns** shelf.
16. Verify that data from both the **Employee** and **Department** datasets is displayed correctly in the worksheet, confirming that the data blending process has been completed successfully.



INSIGHT

- Data from the **HumanResourcesEmployee** and **Department** datasets has been blended successfully.
- The worksheet displays **Department Name** along with **Employee Job Titles**.
- Data blending enables analysis of employee and department information together.



Step 5: Creating Context Filters

1. Open a new worksheet (**Sheet 2**) in Tableau.
2. From the **Data Pane**, drag the **Gender** field and drop it onto the **Filters** shelf.
3. The **Filter [Gender]** dialog box will appear.
4. Under the **General** tab, keep all gender values selected and click **OK**.
5. Right-click the **Gender** filter on the Filters shelf and select **Show Filter**. The **Gender Filter Card** will appear on the right side of the worksheet.
6. Next, drag the **Salaried Flag** field from the Data Pane to the **Filters** shelf.
7. The **Filter [Salaried Flag]** dialog box will appear. Keep the required values selected and click **OK**.
8. Right-click the **Salaried Flag** filter and select **Show Filter**.

9. Now drag the **Hire Date** field from the Data Pane to the **Filters** shelf.
10. The **Filter Field [Hire Date]** dialog box will open. Select **Range of Dates** and click **Next**.
11. Keep the default **Start Date** and **End Date** values, then click **OK**.
12. Right-click the **Hire Date** filter and select **Show Filter**.
13. Verify that all three filters (**Gender**, **Salaried Flag**, and **Hire Date**) are displayed on the worksheet and can be used to interactively filter the data.

INSIGHTS

- Filters allow users to view only the required employee records.
- The **Gender** filter displays employees based on the selected gender.
- The **Salaried Flag** filter helps analyze salaried and non-salaried employees.
- The **Hire Date** filter allows analysis for a specific hiring period.
- Interactive filters make the dashboard more flexible and easier to explore.

Step 6: Creating Bar Chart with Filters (Employees by Department)

1. Open a new worksheet (**Sheet 2**) in Tableau.
 2. Drag the **Name** (Department Name) field from the **Department** dataset to the **Columns** shelf.
 3. Drag the **BusinessEntityID** field from the **HumanResourcesEmployee** dataset to the **Rows** shelf.
 4. Change the aggregation of **BusinessEntityID** to **Count (CNT)**.
 5. Tableau will automatically create a **Bar Chart**.
 6. Rename the worksheet title to "**Employees by Department**".
 7. Drag the **Gender** field to the **Filters** shelf.
 8. In the **Filter [Gender]** dialog box, keep the required gender values selected and click **OK**.
 9. Right-click the **Gender** filter and select **Show Filter** to display it on the worksheet.
 10. Drag the **Salaried Flag** field to the **Filters** shelf.
 11. In the **Filter [Salaried Flag]** dialog box, keep the required values selected and click **OK**.
 12. Right-click the **Salaried Flag** filter and select **Show Filter**.
 13. Drag the **Hire Date** field to the **Filters** shelf.
 14. In the **Filter Field [Hire Date]** dialog box, select **Range of Dates** and click **Next**.
 15. Keep the required date range and click **OK**.
 16. Right-click the **Hire Date** filter and select **Show Filter**.
 17. Verify that the worksheet displays the **Gender**, **Salaried Flag**, and **Hire Date** filters on the right side.
 18. Confirm that the **Employees by Department** bar chart updates interactively whenever any filter value is changed.
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INSIGHT

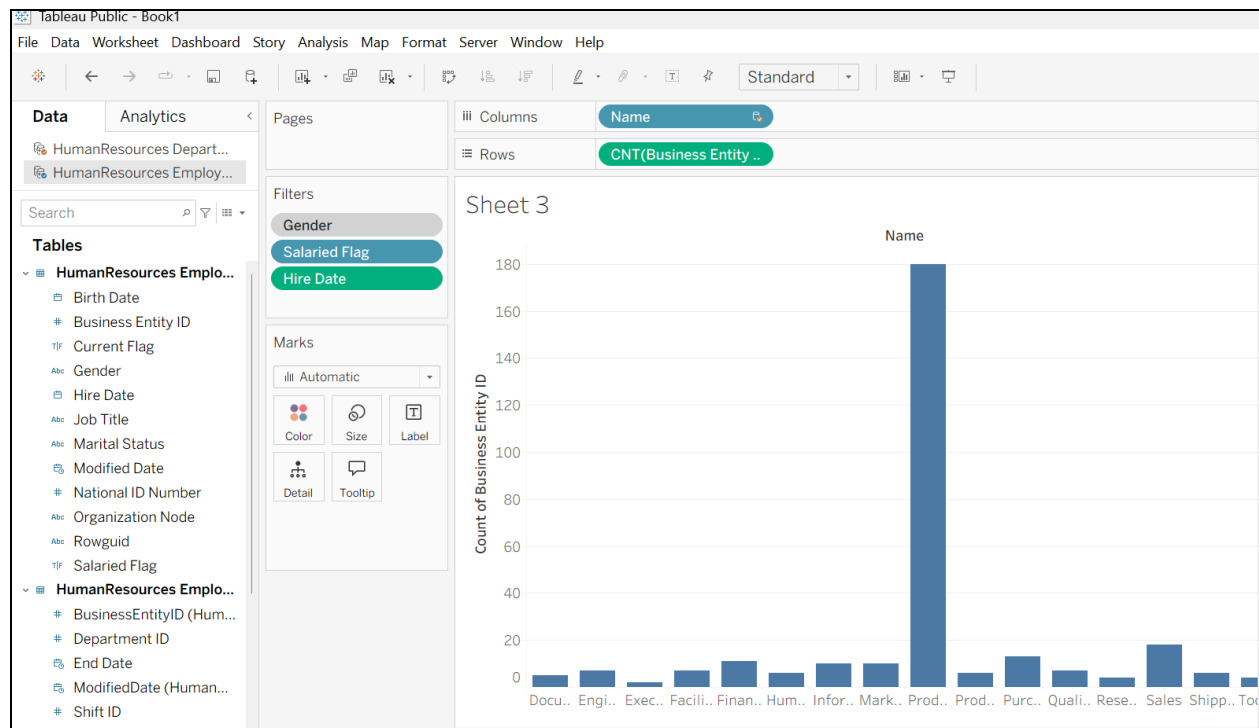
- The bar chart displays the number of employees in each department.
 - Departments with more employees have taller bars, making comparisons easier.
 - Filters help analyze employees based on **Gender**, **Salaried Flag**, and **Hire Date**.
 - The chart updates instantly when filter values are changed.
 - Interactive filters make employee analysis easier and more flexible.
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Step 7: Optimizing Dashboard Performance

1. Click the **Data Source** tab to return to the **Data Source** page.
 2. Ensure that the **Extract** connection mode is selected.
 3. Click the **Edit** option under the **Extract** section.
 4. In the **Extract Filters** dialog box, click **Add Filter**.
 5. Select the **Gender** field and click **OK**.
 6. In the **Filter [Gender]** dialog box, select the required gender (for example, **Male**) and uncheck the remaining values.
 7. Click **OK**, then click **Save Settings** to apply the Extract Filter.
 8. Return to **Sheet 4**.
 9. Right-click the **Gender** filter in the **Filters** shelf.
 10. Select **Add to Context**.
 11. Verify that the **Gender** filter changes to a **gray color**, indicating that it has been added as a **Context Filter**.
 12. Confirm that the worksheet performance is optimized by using both the **Extract Filter** and the **Context Filter**.
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INSIGHT

- The bar chart provides a clear comparison of employee distribution across departments.
- Applying the **Gender** Extract Filter reduces the amount of data processed during analysis.
- The **Gender**, **Salaried Flag**, and **Hire Date** filters allow users to interactively analyze employee records.
- Using a **Context Filter** improves dashboard performance by reducing the amount of data processed before applying other filters.
- Extracts and Context Filters together make the dashboard faster and more efficient.



PRACTICAL INSIGHT

- Different data sources such as **HumanResourcesEmployee**, **HumanResourcesEmployeeDepartmentHistory**, and **Department** can be connected and prepared efficiently in Tableau for analysis.
- Metadata management, joins, and data blending help combine related datasets and improve the quality and completeness of data analysis.
- Dimension, measure, and date filters enable users to explore employee data interactively and focus on specific records based on their requirements.
- The **Employees by Department** bar chart provides a clear comparison of employee distribution across different departments, making workforce analysis easier.
- Using **Extract Filters** and **Context Filters** improves dashboard performance by reducing the amount of data processed while maintaining fast and efficient analysis.